Exercise 9

Grid-based discretized of equation ~~30~~ 29:

Define space and time: $t^n = n\Delta t$ $\quad$ Let $U_i^n = U(x_i, t^n)$

$x_i = j\Delta x$

Given eq 29:

$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x+\Delta x) - 2U(x) + U(x_i - \Delta t)}{\Delta x^2} + O(\Delta x^2)$$

$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x_{i+1}) - 2U(x_i) + U(x_{i-1})}{\Delta x^2} + O(\Delta x^2)$$

$$\left(\frac{\partial^2 U}{\partial x^2}\right)_i^n \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta x^2}$$

Descretized of Eq 33:

$$\frac{\partial U}{\partial z} = \frac{U(z_i + \Delta z) - U(z_i - \Delta z)}{2\Delta z} + O(\Delta z^2)$$

Horizontal velocity U_{ijk}^n, where; i = x-direction index

j = y-direction index

k = z-direction index

n = time index

Discrete approximation: $\left(\frac{\partial U}{\partial z}\right)_{ij,k}^n \approx \frac{U_{i,j,k+1}^n - U_{ijk-1}^n}{2\Delta z}$

with $t^n = n\Delta t$ & $x_i = j\Delta x$

$$\left(\frac{\partial^2 U}{\partial x^2}\right)_i^n \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{2\Delta x^2}$$

$$\left(\frac{\partial U}{\partial z}\right)_{ijk}^n \approx \frac{U_{ijk+1} - U_{ijk-1}}{2\Delta z}$$